

Stochastic Stability via Robustness of Linear Systems

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Time-varying random uncertainty

$$A_k \sim \mathcal{D}$$

is a random matrix drawn from a distribution $\mathcal{D}$ at every time step k

Unifying question

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Are these stability characterizations related?

Prior work:

Robustness $\rightarrow$ Stochastic stability (continuous-time) [1]

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Who cares?

- 1 Answers an interesting open control-theoretic question

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Who cares?

- 1 Answers an interesting open control-theoretic question
- 2 Opens the door to unified analysis & synthesis of controllers for uncertain systems

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Fundamental idea

Restrict search over multiplicative noise distributions to those with

- The same perturbation directions Δ_i
- Standard deviations σ_i related to deterministic robustness margins

Theorem 1

Suppose, for $i = 1, \dots, p$

$$x_{t+1} = (A + \mu_i \Delta_i) x_t$$

is stable for any static bounded perturbation $|\mu_i| \leq \eta_i$ with $\eta_i > 0$.

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Then there exist **computable scalars** $\beta_i \in (0, \infty)$ such that

$$x_{t+1} = \left(A + \sum_{i=1}^p \gamma_{t,i} \Delta_i \right) x_t$$

is ms-stable for any distribution of the independent-across-time random vector $\gamma_t = [\gamma_{t,1} \ \cdots \ \gamma_{t,p}]^\top$ that satisfies

- $\mathbb{E}[\gamma_t] = 0$
- $\mathbb{E}[\gamma_t \gamma_t^\top] = \text{diag}(\sigma_1^2, \dots, \sigma_p^2)$ where $\sigma_i \leq \beta_i$ for $i = 1, \dots, p$.

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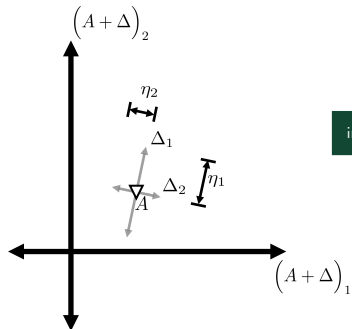
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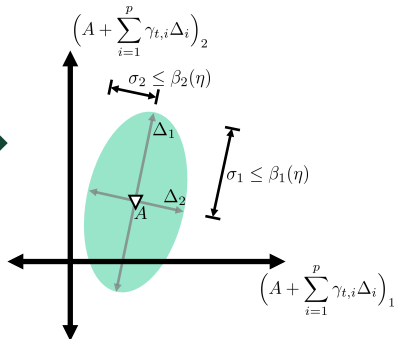
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Proof gives a constructive way to compute ms-stability margins (β_i 's) for the given problem data (A , μ_i 's, Δ_i 's)

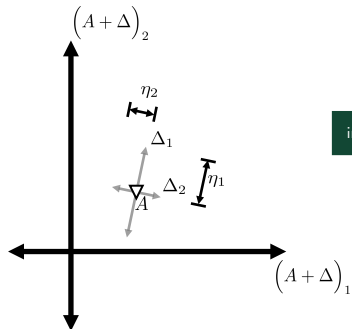
Robust stability



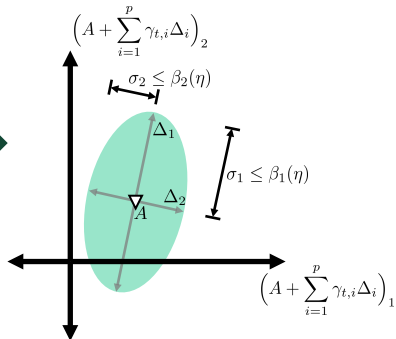
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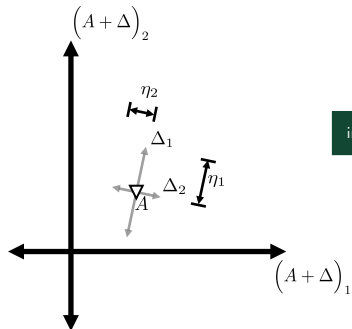


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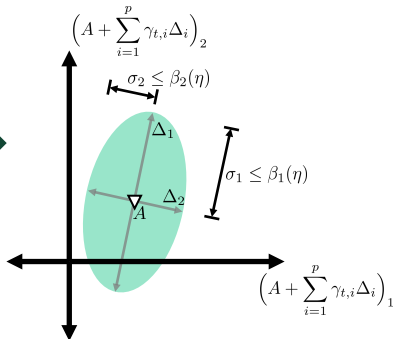


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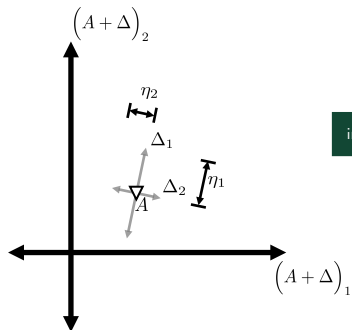
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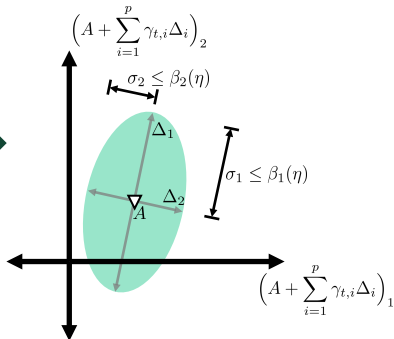
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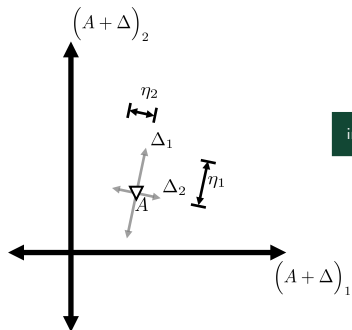
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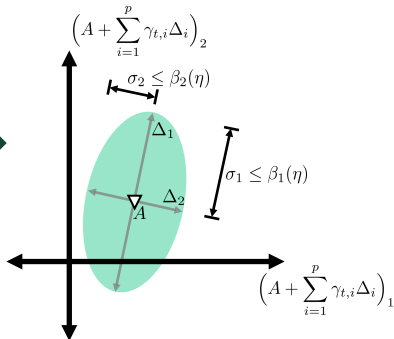
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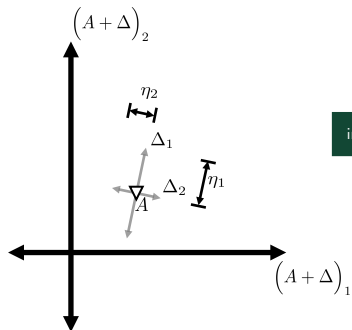
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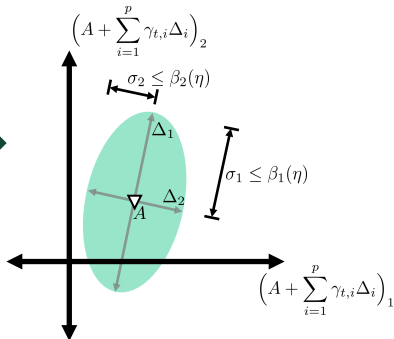
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Remarkably, the claim allows the support of the distribution of stochastic system matrices to be **unbounded**.

Corollary 4

Suppose the semidefinite program (SDP)

$$\begin{aligned} &\text{find} && P \succ 0 \\ &\text{such that} && P \succ \left(A + \sum_{i=1}^p s_i \eta_i \Delta_i \right)^{\top} P \left(A + \sum_{i=1}^p s_i \eta_i \Delta_i \right) \\ &&& \text{for } i = 1, \dots, p, \text{ for any } s_i \in \{1, -1\}. \end{aligned}$$

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- Ms-stability margins increase **linearly** with robustness margins.

How to compute the margins?

Two methods:

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- 1 Solve the **SDP** from Corollary 4

Model of an uncertain rigid pendulum:

$$\underbrace{A = \begin{bmatrix} 0.976 & 0.097 \\ -0.477 & 0.937 \end{bmatrix}}_{\text{Nominal dynamics}}, \quad \underbrace{\Delta_1 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}}_{\text{Uncertain restoring force}}, \quad \underbrace{\Delta_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{\text{Uncertain friction force}},$$

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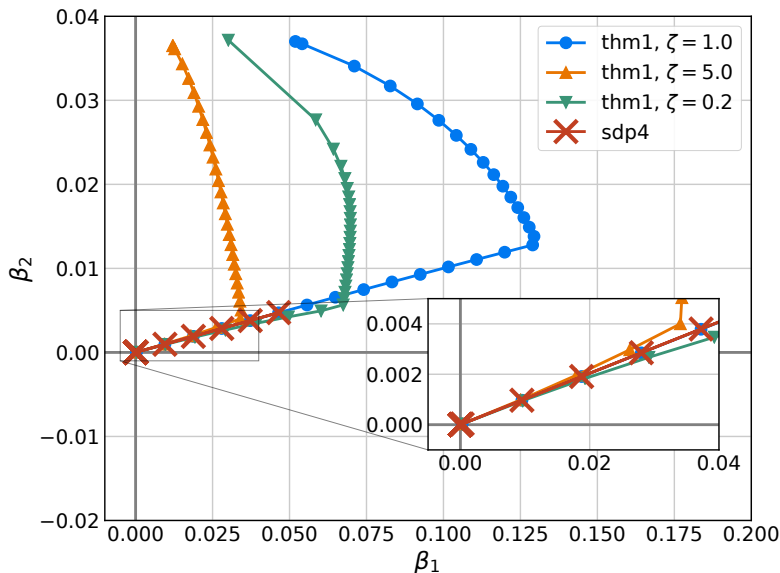
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We also varied a parameter ζ in Algorithm `thm1` to observe its effect.



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is **ms-stable** for any i.i.d. zero-mean random vector γ_t with covariance $\text{diag}(\sigma_1^2, \dots, \sigma_p^2)$ with $\sigma_i \leq \beta_i$.

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- 2 There exist scalars $\beta_i > 0$ such that

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is **ms-stable** for any i.i.d. zero-mean random vector γ_t with covariance $\text{diag}(\sigma_1^2, \dots, \sigma_p^2)$ with $\sigma_i \leq \beta_i$.

Moreover, the margins $\beta_i > 0$ and $\eta_i > 0$ may be computed by SDPs.

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- Extend results to analogous notions for continuous-time systems and nonlinear systems

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